

Suchismith Mahapatra

MACHINE LEARNING/LLM ENGINEER · (MACHINE LEARNING | DEEP LEARNING | NLP)

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About Me

Proficient technical leader with extensive research and developer experience in building custom, robust systems for large-scale learning and driving industry-wide impact. Proven ability of delivering complex technical initiatives from conception to production.

My research interests/expertise lies in:

- ◇ Machine/Deep Learning (ML/DL)
- ◇ Large Language Models:
 - ✓ Pre-training
 - ✓ Preference Optimization (DPO, PPO)
- ◇ Deep Graph/Geometric Learning (GNN)
- ✓ Supervised/Parameter-Efficient Fine-tuning (LORA)
- ✓ Distillation, Quantization (QAT, PTQ)

Skills & Proficiencies

Python | PyTorch | Hive | CUDA | Scala | C/C++ | TensorFlow | LLM inference (vLLM, TensorRT-LLM)

Experience

Meta

Menlo Park, CA

MACHINE LEARNING/LLM ENGINEER

January 2024 - Present

- ◇ Improving YA presence on comment surfaces
 - Built a comprehensive strategy for improving young adults (YA) presence on comment surfaces, resulting in tactical initiatives:
 - + Developed an interesting comments based notification (serving 1.6M users daily) incorporating user personalization in VM, which resulted in strong funnel gains (clicks +0.16%, reactions +0.17%, actions +0.15%).
 - + Built foundational comment-based semantic labels (e.g., p(funny), p(mean), etc.) using Llama-as-oracle which had applications in multitude of products.
- ◇ Llama-as-oracle, summary generation and vibe improvement on comment surfaces
 - Tech lead (4 engineers) for LLM-in-comments work-stream.
 - Built large scale LLM inference pipelines to generate semantic labels for comments using prompt tuned *llama-3.1-70b-instruct*.
 - Utilizing above teacher model, trained highly accurate (both AUC-ROC, PR-AUC ≥ 0.93), “cheap” XLM(R)-based student models for serving billions of comments in production (needed to “optimally” balance compute, latency and performance).
 - Fine-tuned *llama-3.1-8b-instruct* model via SFT + LORA using high quality dataset (1M samples, generated utilizing prompt-tuned *llama-3.1-70b-instruct*) resulting in 75% GPU compute savings and performance gains (+9% PR-AUC, +15% accuracy).
 - Added “guardrail” components and incorporated best practices as part of label generation pipeline to ensure high label quality.
 - Leveraged *llama-4-maverick-17b-128e-instruct* as teacher model in large scale LLM inference pipelines to generate semantic labels for non-english comments from high resource languages.
 - Collaborated with translation team for instruction fine-tuning *llama-3.1-8b-instruct* model to further improve labeling performance.
 - Using above semantic signals in our comment ranking VM boosting funny/interesting comments + demoting bad comments resulted in strong vibe gains (overall vibe +19.2%, vibe “mimicry effect” +1.19%, comment VPV +0.67%, severe bad vibe -27.3%).
 - Built aligned version from *llama-3.1-8b-instruct* base to generate summary for large creators’ posts via RLHF + PPO + LORA and BERT based reward model. Launch resulted in strong user traction (VPV +0.2%, reactions +0.14%). Switched to using simpler DPO strategy.

LinkedIn

Mountain View, CA

SENIOR AI SCIENTIST/ENGINEER

July 2021 - September 2023

- ◇ Conditional generation using LLMs
 - Built LLM prompt generation pipelines for large scale inference to assist in conditional label generation via in-context learning.
 - Instruction fine-tuned *GPT-2-large* base model (774M parameters) to generate summary for members utilizing profile data attributes.
 - Worked towards pre-training in-house BERT model (*LiBERT*) with variety of use-cases.
- ◇ Special Interest Group (SIG)
 - Built a novel unsupervised GNN framework which learns holistic member embeddings via incorporating edge based features in the graph convolution, significantly improved model training speed (-60%) and accuracy (+11%) for clients.
 - Developed a novel strategy for using offline RL methods to build Task-oriented dialogue agents. 📄
- ◇ Standardization/Oribi/Groups
 - Tech Lead for Standardization team (10+ engineers), wherein worked with product managers to convert business/product requirements into practical/scalable technical solutions, applying different ML/DL, GNN and NLP techniques to solve related problems.
 - Led firefighting efforts to quickly resolve P0 issues affecting 725K+ and 183K members which resulted in \$5M+ revenue gain.
 - Improved average coverage of education taxonomy from 74% to 77.2%, which measures to be +5%.
 - Built relevance-based models which significantly improved group post contributions (+19.23%) and consumption (+22.18%).

Amobee

Redwood City, CA

SCIENTIST I

March 2020 - July 2021

- ◊ Novel bidding strategy based on Win Price (WP) estimation
 - Developed and productionized a novel bidding strategy using nonlinear ML based approaches for estimating WP.
- ◊ Factorization Machine (FM/FFM) based ML pipeline for usage in production
 - Led efforts to build a FM/FFM based ML pipeline using a novel sparse matrix formulation that can handle high modality features.
- ◊ BERT/GAN based user embeddings
 - Incorporated BERT/GAN based user embeddings into existing ML/DL models leading to enhanced model accuracy (+7%).

Criteo AI Lab

Palo Alto R&D Center, CA

RESEARCH SCIENTIST

July 2018 - December 2019

- ◊ Improve Click-through and Sales prediction
 - Enhanced existing production Click-through and Sales prediction pipeline using nonlinear ML techniques. Improved stability of our new models significantly from +50% to +5%. A/B test using new models resulted in +3-6% uplift in long-term revenue on all platforms.
- ◊ Theoretical aspects of Deep Learning (worked with [Noureddine El Karoui](#))
 - Working towards understanding kernel and manifold specific aspects of theoretical deep learning.
- ◊ Resolving the posterior-collapse issue in Seq2Seq learning
 - Developed a quantization based approach towards resolving the posterior-collapse issue. 

Criteo AI Lab

Palo Alto R&D Center, CA

RESEARCH SCIENTIST INTERN

May 2017 - December 2017

- ◊ Cross-domain Query-Product (QP) modeling
 - Developed a robust QP model across retailer domains via Domain Adaptation and Optimal Transport based approaches. 

BD Biosciences

San Jose, CA

MACHINE LEARNING ALGORITHM DESIGN INTERN

June 2016 - August 2016

- ◊ Fast Clustering of Flow Cytometry (FC) data
 - Upscaled BD's clustering framework for high dimensional FC data upto ~16x using non-parametric ML techniques.  

Cognizant

Kolkata, India

SOFTWARE ENGINEER

June 2005 - July 2010

- ◊ Developed ExProc, a tool for processing excel documents.
- ◊ Built SuperAgent 4.0, a tool for making reservations which interacts with the Novasol and Cuendet servers.
- ◊ Developed Universal Agent Tool along with my team, a tool which aimed at merging operations for various CRS.
- ◊ Contributed to a white paper Using Venn Diagrams to capture Business Requirements.

Academic Background

University of Buffalo, The State University of New York

Buffalo, NY

PH.D. IN COMPUTER SCIENCE

April 2012 - June 2018

- Topic: Scalable Nonlinear Spectral Dimensionality Reduction methods for streaming data.   
- Advisors: [Varun Chandola](#), [Nils Napp](#) & [Jaroslaw Zola](#) | GPA: 4.0 out of 4.0 ([Transcript](#))

University of Buffalo, The State University of New York

Buffalo, NY

M.S. IN COMPUTER SCIENCE

September 2010 - June 2012

- Topic: A Cold Start Recommendation System Using Item Correlation and User Similarity. 
- Advisor: [Rohini Srihari](#) | GPA: 4.0 out of 4.0 | Department rank: 1 out of 555 ([Transcript](#))

National Institute of Technology, Rourkela

Rourkela, India

B.TECH. IN COMPUTER SCIENCE

August 2001 - May 2005

- Specialization: Discrete Mathematics and Algorithms
- Cumulative Score: 77% (First class with Honors)([Transcript](#)) | Joint Entrance Exam Rank 22 out of 400,000

Honors

2022	Completed NLP / NLU and RL courses as part of AI certification from Stanford University .	Sunnyvale, CA
2022	Was invited to and attended the prestigious 2022 CIFAR DLRL School and OxML 2022 .	Sunnyvale, CA
2021	PC member for ICLR (2021 - present), ACL (2021 - present) and NeurIPS (2021 - present).	Sunnyvale, CA
2020	Was invited to and attended the prestigious Theory of Reinforcement Learning program.	Berkeley, CA
2019	PC member for ICML (2020 - present) and EMNLP 2021 .	Palo Alto, CA
2019	Was invited to and attended the prestigious Foundations of Deep Learning program.	Berkeley, CA

Publications

1. New Methods & Metrics for LFQA tasks. **S. Mahapatra**, [V. Blagojevic](#) and [P. Bertorello](#). 2021 (Preprint available) 
- Interpretable Graph Similarity Computation via Differentiable Optimal Alignment of Node Embeddings. [K. Doan](#), [S. Manchanda](#), **S. Mahapatra** and [C. Reddy](#). Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021 
3. Discretized Bottleneck in VAE: Posterior-Collapse-Free Sequence-to-Sequence Learning. [Y. Zhao](#), [P. Yu](#), **S. Mahapatra**, [Q. Su](#) and [C. Chen](#). 2020 (Preprint available) 
4. Learning Manifolds from Non-stationary Streaming Data. **S. Mahapatra** and [V. Chandola](#). Proceedings of Nature Journal of Big Data, 11(1), 42. 
5. S-Isomap++: Multi Manifold Learning from Streaming Data. **S. Mahapatra** and [V. Chandola](#). Proceedings of 5th IEEE International Conference on Big Data, 2017 
6. Error Metrics for Learning Reliable Manifolds from Streaming Data. **S. Mahapatra**, [F. Schoeneman](#), [V. Chandola](#), [J. Zola](#), [N. Napp](#). Proceedings of SIAM Data Mining Conference, 2017 
7. Modeling Graphs Using a Mixture of Kronecker Models. **S. Mahapatra** and [V. Chandola](#). Proceedings of the 3rd IEEE International Conference on Big Data, 2015. 

Certifications

2023	Building Generative AI Applications 	<i>FourthBrain</i>
2023	Prompt Engineering for LLMs 	<i>Sphere</i>
2022	Designing state-of-the-art Recommender Systems 	<i>Sphere</i>
2022	Natural Language Processing with Transformers 	<i>Hugging Face</i>
2022	Mastering Model Deployment and Inference 	<i>Sphere</i>
2022	Driving Business Impact with Machine Learning 	<i>Sphere</i>
2021	Building Transformer-Based Natural Language Processing Applications 	<i>NVIDIA DLI</i>